

MooMe IoT-AI Livestock Monitoring Platform

The proposed MooMe system is a smart livestock monitoring and automated control platform that uses Internet of Things (IoT) devices and artificial intelligence (AI) to improve dairy farming. The system continuously collects real-time data from each cow and the farm environment to help farmers make better decisions. It is designed not only to monitor cows but also to automatically respond to changes in their health and surroundings. This improves animal welfare, increases milk production, and reduces manual work on the farm.

Each cow is uniquely identified using RFID tags or smart collars, allowing the system to track individual health and productivity. Non-contact temperature sensors are installed above each cow's position to monitor body temperature without disturbing the animal. In addition, environmental sensors such as temperature, humidity, and air quality sensors are installed inside the cowshed. These sensors help detect unhealthy conditions such as high heat or poor ventilation that may affect milk production.

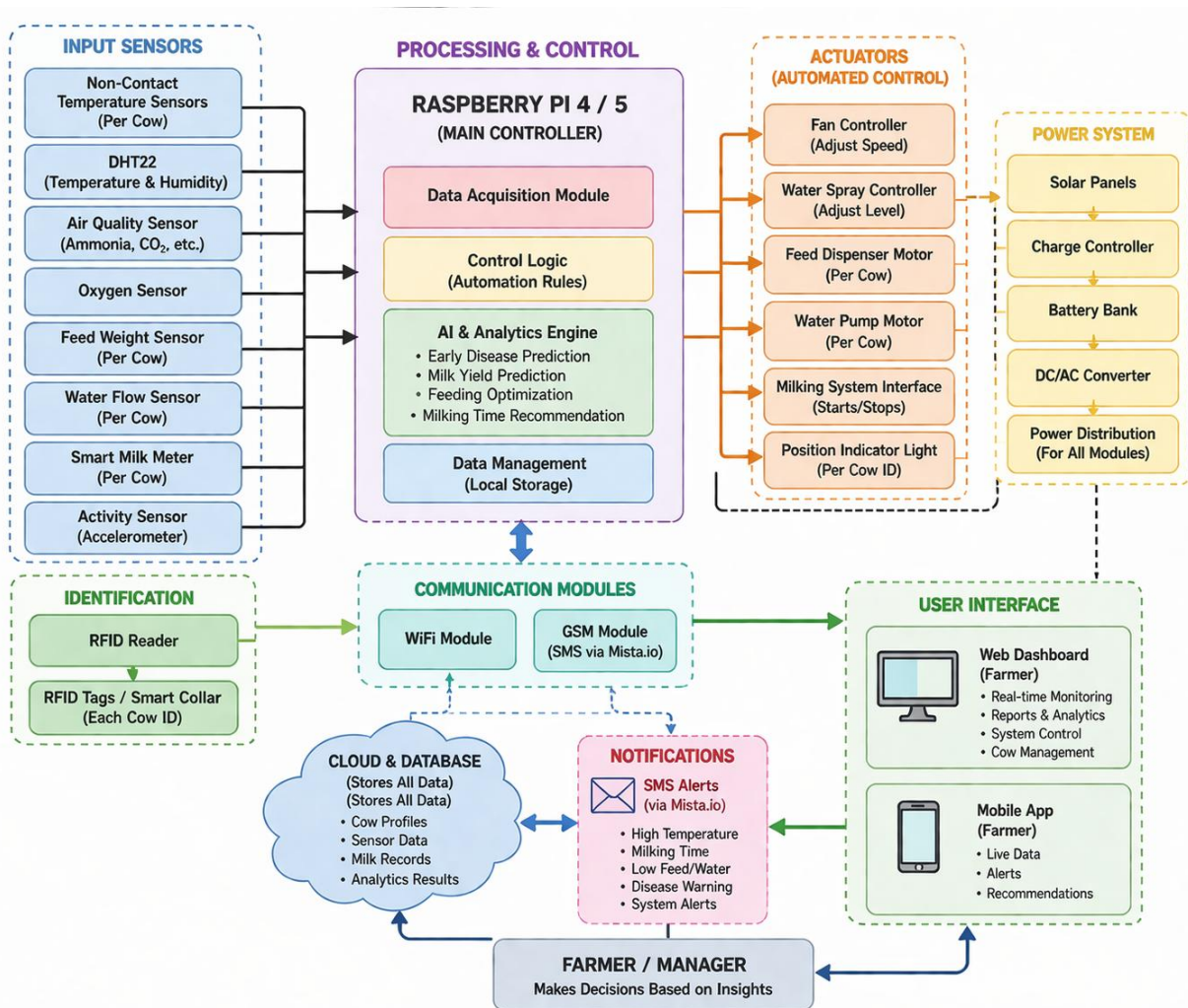
When the system detects abnormal conditions, such as high body temperature, it automatically activates control mechanisms. These include water spraying systems and adjustable fans to cool the cow and maintain a comfortable environment. The system intelligently controls fan speed and water levels based on real-time conditions. This automated response helps prevent stress and disease, which are major causes of reduced milk production.

The system also includes an automated feeding and watering system. Each cow has its own feeding and drinking unit, and the amount of food and water is controlled based on the cow's age and needs, which are set by the farmer through the dashboard. Sensors measure how much each cow eats and drinks per day, and this information is stored for analysis. This ensures that every cow receives the correct nutrition, which directly contributes to improved milk yield.

Milk production is monitored using smart milk meters that record the quantity of milk produced by each cow during milking. The system recommends the best milking times for each cow using AI analysis and notifies farmers through SMS and a dashboard. All data, including milk yield, feeding, and health status, is linked to the cow's unique ID. The system also predicts future milk production and detects early signs of disease, allowing farmers to take action before problems become serious.

Farmers can access all information through a mobile or web-based dashboard, where they receive real-time updates, alerts, and recommendations. SMS notifications are sent using platforms like Mista.io while GSM will be used to ensure communication even without internet access. The system is powered by a solar backup to ensure continuous operation in areas without reliable electricity. Overall, the MooMe system provides a complete smart farming solution that combines monitoring, automation, and AI-driven decision-making to increase productivity and efficiency in dairy farming.

Block Diagram of system



MooMe System – Cost Estimation (12 Cows)

No	Component	Description	Quantity	Unit Price (RWF)	Total (RWF)
1	Raspberry Pi 4/5	Main controller	1	120,000	120,000
2	MicroSD Card (64GB)	Storage	1	15,000	15,000
3	Power Supply	Adapter	1	10,000	10,000
4	Non-contact Temp Sensor (IR)	Per cow	12	20,000	240,000
5	DHT22 Sensor	Environment monitoring	2	8,000	16,000
6	Air Quality Sensor (MQ-135)	Gas detection	1	15,000	15,000
7	Oxygen Sensor	Air monitoring	1	25,000	25,000
8	RFID Reader Module	Identification	1	18,000	18,000
9	RFID Tags	Per cow ID	12	2,000	24,000
10	Water Flow Sensor	Per cow intake	12	10,000	120,000
11	Load Cell + HX711	Feed measurement	12	15,000	180,000
12	Smart Milk Meter	Milk measurement	12	60,000	720,000
13	DC Water Pump	Per cow	12	20,000	240,000
14	Fans (Cooling)	Shared system	4	25,000	100,000
15	Relay Modules	Control system	5	8,000	40,000
16	Motor (Feed Dispenser)	Per cow	12	18,000	216,000
17	Pipes & Spray Nozzles	Cooling system	1 set	60,000	60,000
18	GSM Module (SIM800L)	SMS alerts	1	15,000	15,000
19	Wi-Fi Router	Connectivity	1	35,000	35,000
20	Solar Panel (200W)	Backup power	1	150,000	150,000
21	Battery (100Ah)	Energy storage	1	180,000	180,000
22	Charge Controller	Solar system	1	40,000	40,000
23	Inverter	Power conversion	1	80,000	80,000
24	Wires & Installation	Setup	1	80,000	80,000

The total cost for all components (12 cow's system) is:

₹ Total = 2,699,000 RWF

Its estimation cost may increase or decrease

System Architecture

